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When Growth Detaches from Labor: How AI Disrupts the Macroeconomic Framework

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Executive Summary: The Great Decoupling

This paper is a philosophical exploration of a potential future rather than a report on current economic conditions. It serves as a **framework for thought** regarding the ways the world may change as we enter an Agentic era. While our current system remains intact today, I believe the modern macroeconomic operating system is approaching a point of structural failure. For over a century, productivity gains supported human labor. However, a transition toward a world driven by artificial intelligence suggests that output could eventually scale through compute rather than human headcount. If these views hold true, the foundational links driving models like the Phillips Curve and Okun's Law will fracture as growth begins to detach from labor participation. This gets away from the debate of job destruction and focuses on the relationship between labor and our historically based thinking around macroeconomics in a world of digital knowledge workers and physical humanoid laborers.

For investors, this shift suggests a move away from cyclical timing toward a practice of structural handicapping. In this potential future, AI innovation could compress decades of creative destruction into months. This would create a two tier economy defined by massive capital concentration in physical infrastructure like energy and chips alongside hyper accelerated churn in software applications. As corporate decisions shift from the cost of hiring to the cost of compute, traditional monetary policy may lose its efficacy. In such a regime, rigor is not found in the precision of a back tested model but in the ability to update probabilities as the exponential curve of intelligence moves from scarcity toward abundance.

Investor Implications

If growth begins to detach from labor and macroeconomic relationships become less stable, several structural shifts for investors are likely to follow:

- Higher volatility and higher volatility of volatility as markets adjust to faster technological change and shorter competitive cycles.
- Wider credit spreads as earnings visibility declines and fixed obligations become riskier in a world of accelerating disruption.
- Long duration assets face structural pressure as predictability declines and discount rates reflect greater uncertainty about the future.
- Liquidity becomes increasingly valuable as investors favor assets that can adapt quickly in a rapidly changing economic environment.
- Passive investment strategies face growing challenges because capitalization weighted indexes allocate capital based on past winners rather than emerging leadership.

- Equity dispersion increases as innovation cycles accelerate, creating larger gaps between companies that successfully adapt to AI and those that do not.
- Real assets tied to energy, compute, and physical infrastructure gain strategic importance as AI development requires massive increases in power, chips, and data centers.
- Corporate leverage becomes more dangerous as faster creative destruction shortens the lifespan of competitive advantages.
- Monetary policy becomes less predictive as traditional transmission channels tied to labor markets weaken.
- Liquidity conditions become a more important driver of markets than traditional macro indicators such as unemployment or GDP.

The ultimate endgame described here is a shift toward fiscal dominance and a total revaluation of the capital structure. If the labor wage consumption cycle weakens as I expect, political pressure will likely force even greater fiscal intervention and a reorganization of the tax base. In this scenario, sovereign debt might lose its status as a safe asset and begin to carry equity like risk. This paper provides a framework for navigating this macro paradox. It is an argument that the most successful investors in the coming decade will be those who stop trying to fit an exponential future into a linear past and instead prepare for a world that does not yet exist.

It also opens the door for a better understanding very soon for all investors around how the financial guardrails of the world will also change to support the Agentic world and a world of deepfakes and cyber-attacks.

A Breakfast Discussion

First of all, this is the first long form philosophical piece I have written for 22V. I usually save this for Substack but this one is something I have worked on for a long time and I think it has become more important due to how fast things are moving and the disruptions already taking place, so the time for posting it is now. It is long but you can always put it into your favorite LLM if you want a summary beyond the Executive Summary, but I do think it is an important one for all investors and parents to think about as we accelerate in the year of the rise of the Agentic world.

This week I sat down for breakfast to share my current views, which triggered finishing the paper on the train. One of the people at the table was a smart, experienced, well read person who thinks in macroeconomic terms. He has a deep understanding of the ways the macro economy has worked historically and strong beliefs about what has driven growth.

I have historically loved these back-and-forth discussions but I have learned over the last year in these types of conversations that we will inevitably reach the point where my views around the singularity and what the future may look like lead to a deep difference in opinion. Once the question at the table becomes, "What do you think the world will look like in five to ten years," it always seems to bring out emotions.

For those wondering why I write things for 22V but also on Substack, it is this reason. I want to be able to share my deep philosophical views ten years out regarding AI, jobs, the economy, and especially crypto, because none of us know but I do think about it a lot and what I write today will inevitably change along the way. Most importantly, you cannot make money off ten years from now, so I would never send something to an audience of 22V investors making decisions each day that does not help that process.

For this five-to-ten-year view, I described a world where artificial intelligence replaces not just tasks but entire cognitive functions at scale. Where humanoid robots begin entering the physical labor force. Where the speed of innovation compresses what used to take decades into years. Where the macroeconomic frameworks we were both trained on, the ones built on labor participation, wage transmission, and consumption driven growth, begin to fracture under the weight of a technology that does not need people to produce output.

The reaction was immediate and familiar. Strong disagreement. Too much narrative, not enough facts. Where are the models? Where is the data? You are telling a story, not making an argument.

All true when pushed into five-to-ten-year views, and what makes it worse is that I said it with conviction but not certainty since we do not know. When sharing views, they should come with conviction or not at all in my opinion. Make your own decisions but collect anyone who has thought deeply about an important topic. Since my visit to Singularity University in 2013, this has been my focus, not since AI became a headline.

This back and forth about facts versus things that are difficult to measure, like the future, is a divide that has defined much of my career. Equities versus bonds, or hope versus facts. Discretionary versus quantitative. People who think in narratives about the future versus people who think in models built on the past. And it only gets worse if I bring up Bitcoin, which is why I have learned to draw the line at certain tables.

But here is what I have come to understand. Like everyone else, I do not know for certain what the future holds. I am not making predictions with false precision. I am a handicapper.

I grew up in a world of probabilistic thinking, learning from my father at the racetrack that the goal is never certainty. The goal is to identify where the odds are mispriced and adjust as new information arrives. When you meet me, never mistake my conviction for certainty. My father told me when I was very young, do not believe anything anyone says. Use it in your own decision making but do not believe it as fact. That applies to everything I write or say here. This is how I think the journey of AI will play out for macroeconomics and the capital structure.

What has changed is the speed at which new information arrives. The AI acceleration I write about constantly is not just transforming industries. It is compressing the timeline of change itself. The future is closer than any linear model suggests. And that makes timing, the one thing every macro strategist cares about, nearly impossible to calibrate using traditional frameworks.

I live in a Bayesian world of shifting probabilities. I always have. But the rate of Bayesian updating has never been this fast. Every week brings new model capabilities, new robotic demonstrations, and new infrastructure buildouts that looked like science fiction eighteen months ago. I am not extrapolating a trend line. I am watching an exponential curve that is replacing human intelligence and will soon replace human movement and physical labor through humanoid robotics.

We are entering a world that is genuinely impossible to predict with precision, but the direction is clear to me. AI will break the macroeconomic and capital structure frameworks that the modern financial system was built on. Not because of a narrative. Because when you can produce output without capital and without employees, and you can do it at a speed that compresses competitive cycles into months, the assumptions underneath every macro model we use stop holding.

We just witnessed a sixteen-year period where the Mag 7 grew to dominate global equity markets and did not use debt to get there. Over the last year we saw GDP growth without job creation. Now we are seeing long duration assets fall under the uncertainty of what the world might look like three years from now.

This paper is my attempt to lay out why. Not as a prediction with a date attached, but as a framework for thinking about what happens when growth detaches from labor, when intelligence becomes abundant, and when the macroeconomic architecture of the last century meets a force it was never designed to accommodate.

Given the viral Citrini paper and the subsequent reaction, along with the massive dispersion happening in the market, we have moved past whether AI is overhyped and into the more challenging territory: how do we adapt as investors.

The Industrial Growth Framework: Labor as the Anchor

From the Industrial Revolution through the digital age, economic growth followed a recognizable pattern. Productivity improvements, whether steam power, electricity, assembly lines, or computing, amplified human labor. Workers became more productive. Firms scaled. Wages rose. Consumption expanded. The middle class grew alongside industrial capacity.

The foundational growth equation reflects this:

Growth = Population Growth + Capital Accumulation + Productivity Gains

Crucially, productivity gains were largely complementary to labor. Machines enhanced human output rather than replacing cognition itself. Even when automation displaced certain jobs, new industries emerged that reabsorbed workers. The system self corrected.

Financial assets evolved as claims on this compounding process. Equity valuations were justified by rising earnings from expanding markets. Government bonds were backed by growing tax revenues from broad income gains. Fiat currencies represented trust in expanding economic capacity.

Every major macroeconomic framework of the twentieth century was built on this architecture. The Phillips Curve assumed a stable relationship between unemployment and inflation. Okun's Law linked GDP growth to employment changes. The natural rate of interest derived from assumptions about productivity and labor dynamics. Fiscal multipliers assumed government spending circulated through wage earners who spent locally. This was not just a model. It was the operating system of modern capitalism, and it worked, more or less, for a hundred and fifty years.

AI Breaks the Labor-Output Link

I believe if you have managed people for a long time and are using AI all day, I don't think you can have any other view based on my own experience. Artificial intelligence challenges this labor output foundation at its core.

However, given the debate on the recent Citrini piece, I do want to make it clear that I do not believe in a world where unemployment goes higher. Adoption will be more challenging than expected, both culturally and structurally, given legal and compliance friction at larger enterprises. My belief is in the hyper growth of entrepreneurs competing with slow moving, bloated organizations and their AI committees, puzzles of unstructured data, and surfaces ripe for cyber-attacks.

Unlike prior technological waves, AI does not simply amplify physical or clerical work. It replicates cognitive functions: pattern recognition, language generation, decision making, and increasingly, reasoning. For the first time in economic history, the bottleneck of human intelligence can be partially, and in many domains fully, automated. This introduces a dynamic with no precedent in macroeconomic thought:

Output can scale without proportional labor input.

A startup can achieve global reach with a fraction of the workforce historically required. We have seen this in the rapid rise of revenue per employee at startups reaching milestone growth in record time. Layers of white collar employment, including legal review, financial analysis, customer service, content production, and software engineering, can be compressed or eliminated. Nominal world GDP is approximately 120 trillion dollars. Wages account for roughly 60 trillion. That is an enormous amount of productivity that can be increased at the expense of labor.

The traditional transmission mechanism:

More Workers – More Output – More Income – More Consumption

begins to shift toward one of two outcomes:

More Compute – More Output – More Profit Concentration

or:

More Compute – More Output – More Democratization of Profits

I lean towards the latter but understand the argument for the former.

And with the rapid advance of humanoid robotics, in a not-so-distant future, this will not be limited to cognitive work. The physical labor force, warehousing, manufacturing, logistics, and eventually services, faces the same structural displacement. When intelligence and physical capability are both automatable, the labor anchor that held the macroeconomic model in place loses its grip entirely. This is not a cyclical disruption. It is a regime change in how output relates to human participation.

When Intelligence Becomes Abundant

In the industrial era, capital and labor were both scarce. Human intelligence was constrained by time, education, and experience. Knowledge accumulated slowly across careers and generations. This scarcity gave labor its value and its central place in economic models.

AI shifts intelligence from scarcity toward abundance. Models replicate knowledge instantly at near zero marginal cost. Once trained, they scale globally without additional labor input.

This is the mechanism driving everything else in this paper. When intelligence becomes abundant, the marginal value of individual cognitive labor declines, competitive advantages built on knowledge become fragile, information asymmetries compress, barriers to entry collapse in knowledge intensive industries, and the speed of competition accelerates beyond human reaction time.

If intelligence is no longer the scarce factor, what is? The answer shifts toward physical infrastructure, energy access, data ownership, regulatory positioning, and network coordination. Value creation migrates from labor intensive firms toward capital intensive platforms and physical bottlenecks.

Macroeconomic models have no framework for an economy where the historically scarce input, human cognition, suddenly becomes abundant. Every assumption about growth decomposition, labor markets, and productivity measurement was built for a world where intelligence was the binding constraint. Removing that constraint does not improve the model. It breaks it.

The Macroeconomic Frameworks That Break

The Solow Residual Swallows the Model

The Solow model separates GDP growth into contributions from labor, capital, and a residual called total factor productivity. TFP has always been the least understood component, a catch all for technological progress and anything else not directly attributable to labor or capital.

AI threatens to make TFP the dominant term. If artificial intelligence drives the majority of productivity improvement while labor contributes less and capital concentrates, the decomposition becomes lopsided. The residual swallows the model. If the thing measured least precisely becomes the primary driver of growth, our ability to forecast and set policy using traditional frameworks degrades substantially.

The Phillips Curve

The Phillips Curve posits a stable tradeoff between inflation and unemployment. Central banks have used it as a cornerstone of monetary policy for decades. AI disrupts this directly. If output scales without proportional hiring, economic growth can coexist with stable or rising unemployment. Deflationary pressure from AI driven productivity can occur simultaneously with labor market weakness. The curve does not just flatten. It potentially inverts or becomes meaningless as a policy guide.

Okun's Law

Okun's Law links GDP changes to unemployment changes. If AI enables GDP growth without proportional job creation, the relationship decouples. Growth and employment no longer move in tandem. Policymakers using GDP as a proxy for labor market health find the signal increasingly unreliable.

The Natural Rate of Interest

R-star depends on assumptions about productivity growth, demographics, and savings investment dynamics. AI scrambles every input. If productivity accelerates but returns concentrate in capital rather than distributing through wages, does the neutral rate rise or fall? The frameworks used to estimate r-star were not built for an economy where productivity

and wage growth decouple. Operating monetary policy around an already uncertain estimate becomes substantially more dangerous when the structural relationships underneath it are in flux.

GDP as a Measure of Welfare

GDP becomes unreliable as a welfare measure in an AI driven economy. If a model generates in seconds what used to require a team for weeks, where does that value appear in national accounts? Conversely, GDP could rise while median living standards stagnate, with welfare gains concentrating among capital owners. The dashboard says everything is fine while the engine runs on a fundamentally different fuel.

The Monetary Transmission Mechanism Fractures

Monetary policy works through credit channels that depend on a functioning link between interest rates and real economic activity, primarily through labor markets and wage sensitive consumption. If fewer people earn wages tied to output, that mechanism weakens at its source. Rate cuts stimulate borrowing and spending, but if households face structural displacement rather than cyclical unemployment, cheaper credit does not solve the problem.

On the corporate side, AI driven firms may be less sensitive to rate changes. Companies generating output through compute rather than labor have different cost structures. Their investment decisions are driven by model capability and infrastructure buildout, not the marginal cost of hiring. Rate changes may move financial asset prices without proportionally affecting real economic decisions.

This creates a regime where monetary policy becomes primarily a tool of financial asset management rather than real economic stabilization. We may have already seen the early stages of this dynamic in the post 2008 era, where aggressive monetary policy lifted equity markets without producing commensurate wage growth. AI accelerates and deepens this disconnect.

Fiscal Multipliers Compress

Government spending historically multiplied through labor intensive, domestically consumed channels. If AI changes the production function so that fewer workers fulfill government contracts, or if spending flows to capital intensive AI infrastructure providers employing relatively few people, the multiplier shrinks.

Consider a government infrastructure program. Traditionally, thousands of construction workers, engineers, and suppliers. In an AI and robotics augmented economy, a fraction of the workforce, with more value flowing to equipment manufacturers, software providers, and a smaller number of highly skilled operators. The transmission from government spending to broad economic participation becomes less efficient. Policymakers estimating impact using historical multipliers may systematically overestimate employment and consumption effects.

Growth Without Wage Distribution

This is the central distributional consequence. In the industrial era, rising productivity supported rising wages over long horizons. Consumption demand increased in parallel with corporate earnings. The system was self reinforcing.

Labor's share of GDP has already been declining since the early 2000s, well before the current AI wave. As this share has declined it should not be surprising that Government debt to GDP has risen as the fiscal dominance has taken over the fill the gap. This is also seen in the K Shaped economy. AI does not create this trend. It accelerates it dramatically. As AI handles increasing portions of cognitive and eventually physical labor, corporate profits rise, labor's share falls further, middle skill employment shrinks, and wage growth stagnates for large segments of the population. GDP can rise while median household income does not follow.

This creates a macro paradox. Aggregate demand depends on broad purchasing power. If wages stagnate while production scales, who buys the output? The industrial model solved this through the wage consumption cycle. An AI driven economy may not. Meanwhile, the composition of savings and investment reorganizes. Corporate savings dominate household savings. Investment concentrates in AI infrastructure and compute rather than labor intensive production. The banking system's traditional role of intermediating between household savers and business borrowers

changes character as financial markets increasingly intermediate between concentrated corporate capital and infrastructure deployment.

Accelerated Creative Destruction

AI changes the speed of competition, not just the level of output. Industry cycles compress. Competitive advantages erode faster. Software moats narrow as models commoditize functionality. Entire business models face disruption within years rather than decades. This is why I reference Joseph Schumpeter so much in my writings.

Fiat financial assets rely on predictability: predictable earnings, predictable tax bases, predictable growth trajectories. When creative destruction accelerates, forecast horizons shorten, earnings visibility declines, equity risk premiums rise, and debt becomes riskier as stable cash flows grow scarcer. Government debt is not immune. If AI concentrates returns and increases economic volatility, the tax base becomes less stable. Debt that once appeared safe begins carrying equity like risk as the underlying structure becomes unpredictable.

Concentration and the Paradox of AI Capital

AI produces a paradox in capital allocation. Large scale models to get higher and higher IQs for the models require massive infrastructure: advanced semiconductors, high bandwidth memory, data centers, and enormous energy capacity. Capital concentrates toward these layers. Simultaneously, application layer businesses face rapid commoditization. The barrier to entry for knowledge intensive businesses drops while the barrier for infrastructure rises.

This produces a two tier economy. Massive capital concentration at the top and accelerating churn below, with increasing dispersion within equity markets. The fiat system historically benefited from broad corporate participation in growth. AI may instead produce narrower profit pools at the top combined with higher churn among mid tier firms. When returns concentrate this sharply, the tax base narrows, political pressure intensifies, and the feedback loops that sustained broad based growth weaken. This is not just a market structure issue. It is a macroeconomic stability issue.

The Political and Fiscal Regime That Follows

If AI weakens wage growth while increasing capital concentration, innovation becomes funded by concentrated corporate profits and sovereign capital rather than broad middle class expansion. Political pressure follows inevitably. Labor displacement at the scale AI implies does not happen without political response: greater fiscal intervention, industrial policy targeting employment, regulation of AI concentration, new tax structures targeting capital and compute, and experiments with universal basic income.

Fiscal dominance may become the default regime. If monetary policy loses transmission effectiveness because the labor channel weakens, governments rely increasingly on direct fiscal intervention to maintain demand. Deficits expand. Central bank independence erodes. Fiat currencies no longer represent just economic growth but political management of technological disruption. The value of a currency becomes a function of how effectively a government manages the transition from a labor based to a machine-based economy.

The implications for sovereign debt are acute. If governments must spend more while their tax bases narrow and grow more volatile, debt sustainability calculations change. The arithmetic that made sovereign debt safe in a world of steady GDP growth and broad tax bases may not hold in a world of concentrated AI driven output and structurally weaker labor participation.

For macroeconomists who model economies as systems with stable structural relationships, this is deeply uncomfortable. The models lose internal consistency as economic relationships decouple and become dependent on political inputs that resist quantification.

The Psychological Fracture

Perhaps the deepest disruption is psychological. Of everything I covered, my strongest convictions lies in the pace and disruption of AI in coming years and the psychological damage it will cause over the next five years. Humans do not like

change. For generations, people equated growth with human effort. Work was central to identity and economic participation. AI decouples growth from human labor, shifting the role of human work from central to conditional.

Consumer confidence depends on people feeling they participate in growth. Business confidence depends on manageable competitive dynamics. Policy confidence depends on frameworks that explain and predict. When all three loosen simultaneously, macro instability can emerge from sentiment shifts no model anticipates. The behavioral assumptions embedded in macroeconomic frameworks, rational expectations, stable preferences, predictable responses to incentives, all rest on a world where people understand their role in the economy. Dislocate that understanding, and the models lose not just their structural relationships but their behavioral foundations.

Two Possible Paths

Path One: Distributed Abundance. AI dramatically lowers costs. New industries emerge around human AI collaboration, creative work, and experiences that resist automation. Entrepreneurship flourishes at lower capital thresholds. Energy becomes abundant through AI optimized systems. Productivity gains translate into lower costs of living that offset wage stagnation. The macro frameworks adjust, bruised but functional.

Path Two: Concentrated Acceleration. AI concentrates returns in capital intensive infrastructure. Labor participation weakens structurally. The speed of disruption outpaces institutional adaptation. Asset prices decouple from median income. Debt structures face instability. The macro frameworks fail as predictive or policy tools.

The truth will involve elements of both. But the range of outcomes is wider than anything the post war macroeconomic consensus was built to handle. And the speed, the dimension my breakfast companion found most difficult to accept, determines which path dominates in what timeframe. Maybe it is the optimist in me but I believe in Path One.

The Handicapper's View

I return to where I started. I am not a macro theorist. I am a handicapper who learned probability at the racetrack and spent thirty years applying it on Wall Street. I do not pretend to know the future with precision. No one does, and anyone who claims otherwise in the face of exponential technological change is more confident than the situation warrants.

But I can read the odds. And the odds are shifting faster than at any point in my career. Every week, the probability distribution updates. New AI capabilities. New robotics demonstrations. New infrastructure buildouts. The Bayesian updating is relentless, and it consistently shifts probability mass toward outcomes that would have seemed implausible two years ago.

The macro community is not wrong to demand rigor. But rigor built on frameworks that assume stable structural relationships is not rigor when those relationships are breaking. It is precision applied to the wrong model. When the underlying structure of an economy is shifting, the most rigorous thing you can do is acknowledge the shift and reason carefully about its implications rather than insisting on models whose assumptions no longer hold.

The macroeconomic frameworks we inherited, the Phillips Curve, Okun's Law, the Solow decomposition, standard monetary transmission, historical fiscal multipliers, were built for a world where human intelligence was scarce and labor was central to output. They served that world well. But that world is ending.

What replaces it is not yet clear. That is the honest answer. But the direction of travel is unmistakable. Growth is detaching from labor. Intelligence is becoming abundant. The speed of change is accelerating. And the macroeconomic architecture of the last century is not equipped to handle what comes next.

The question for investors, policymakers, and economists is whether these frameworks adapt in time, or whether we navigate the most consequential economic transformation in modern history using maps drawn for a world that no longer exists.

I know which side of that bet I am on. And I know the odds are shifting every day.

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